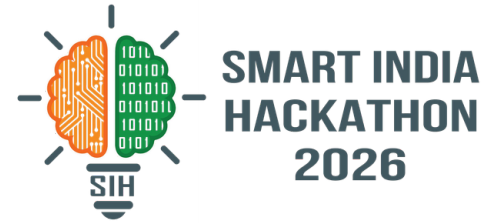


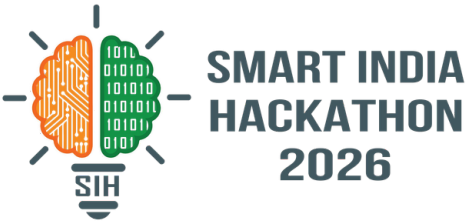
SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – SIH26018**
- **Problem Statement Title-Intelligent Land Record Digitization and Validation System**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-**
- **Team Name - Nexora**





IDEA TITLE

FOR REVENUE OFFICERS

1. SMART DOCUMENT INTAKE

- Upload scanned PDFs, images and handwritten land records
- Batch processing for multiple pages/documents
- Automatic image cleanup: deskew, denoise and contrast enhancement

2. STRUCTURED DIGITIZATION

- Multilingual OCR for Kannada, English, Hindi and extensible Indian languages
- AI extracts owner, survey/khasra, khata, area, village, taluk, district and land details
- Converts unstructured documents into searchable digital records

3. VALIDATION & RISK REVIEW

- Business-rule and reference-record validation
- Detects field conflicts, inconsistent values and possible duplicates
- Generates field-level and overall confidence scores

4. HUMAN VERIFICATION

- Side-by-side original document and extracted data
- Officer can edit, approve or reject uncertain records
- Every correction is recorded for audit and future model improvement

AI-POWERED PROCESS

MULTILINGUAL OCR

- Extract text from faded, noisy and handwritten records
- Preserve page/document structure for downstream AI

AI FIELD EXTRACTION

- Map OCR text into standardized land-record fields
- Handle variations in names, numbers and document formats

VALIDATION + DUPLICATE ENGINE

- Compare extracted values with reference/mock records
- Flag mismatches, duplicates and suspicious inconsistencies

TRUST SCORE + EXPLAINABILITY

- Explain why a record is flagged
- Prioritize low-confidence/high-risk records for officers

INNOVATION & UNIQUENESS

LAND RECORD TRUST SCORE

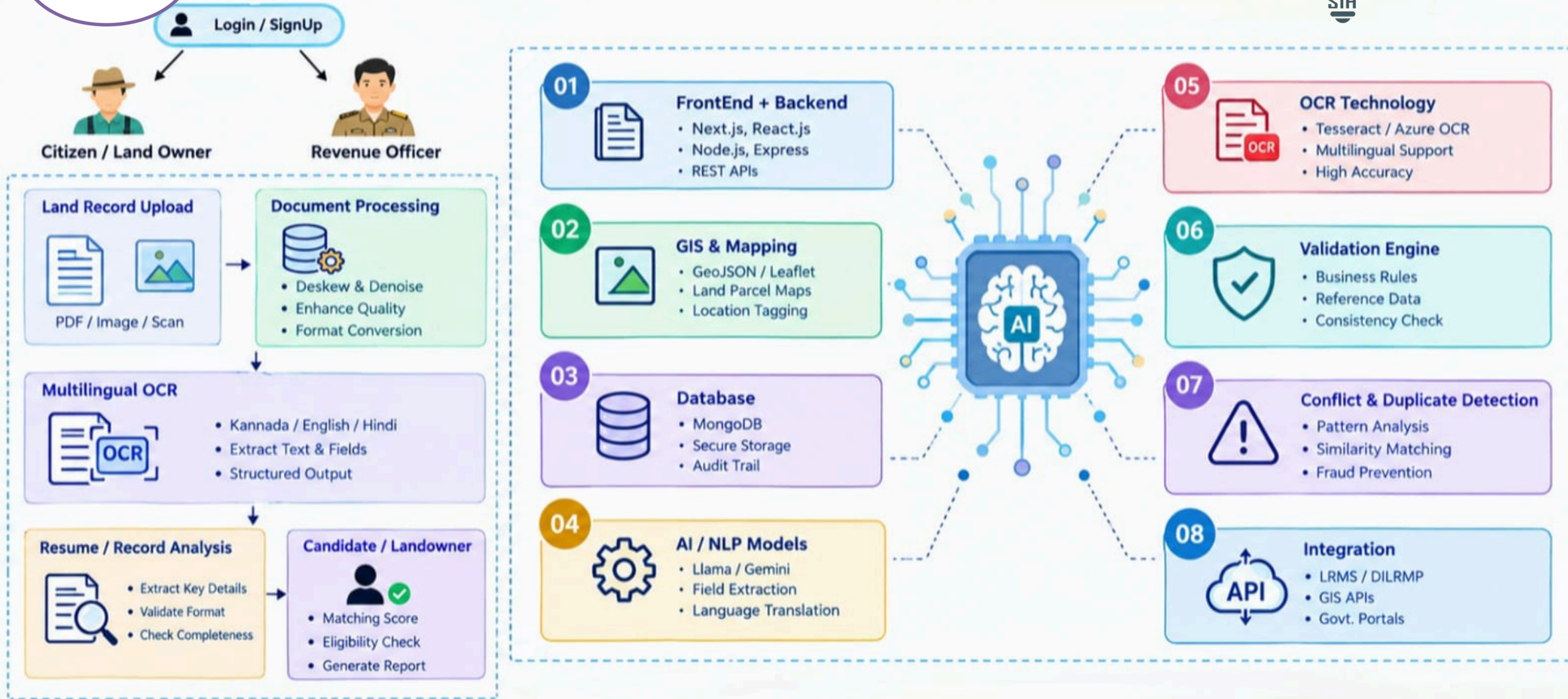
Explainable 0–100 reliability score using OCR confidence, validation, conflicts and duplicate risk.

CONFLICT EXPLANATION

Shows exactly which fields disagree and why the record needs human review.

INTEGRATION + GIS READY

APIs and map-ready structured data support future LRMS/DILRMP/GIS integration.





**01**

Faster Digitization

- Converts old scanned and handwritten records into structured digital data
- Reduces manual data-entry workload

**02**

Improved Accuracy

- AI-powered extraction with confidence scoring
- Detects missing, inconsistent and conflicting information

**03**

Reduced Fraud & Duplication

- Identifies duplicate records and suspicious inconsistencies
- Supports reliable land-record verification

**04**

Better Decision Making

- Dashboard provides real-time record status and conflict insights
- GIS visualization helps authorities identify problematic records

**05**

Transparency & Accountability

- Human verification keeps officers in the loop
- Audit trails track corrections and approvals

**06**

Scalable Digital Governance

- Multilingual and API-ready architecture
- Can integrate with LRMS, DILRMP and GIS systems

1

**Department of Land Resources – DILRMP**

Government of India's Digital India Land Records Modernization Programme focuses on digitization, modernization and integration of land records.

2

**DILRMP 3.0 Operational Guidelines (2026–2031)**

Current framework for modernizing land records and improving digital land administration.

3

**DILRMP-MIS 3.0**

Information on computerized land records, cadastral map digitization and integration of land-record systems.

4

**Open Government Data (OGD) Platform India**

Government datasets and progress related to land-record modernization.

5

**Research Areas**

- Optical Character Recognition (OCR)
- Multilingual Indian-language document processing
- Natural Language Processing (NLP)
- AI-based information extraction
- Data validation
- Duplicate detection
- GIS and spatial data integration
- Human-in-the-loop verification

6

**Technologies Referenced**

- Python & OpenCV for document preprocessing
- OCR for text recognition
- NLP/LLM for structured field extraction
- Database systems for digital record storage
- GIS for map-based visualization and verification